

Finding an edge of a cuboid

The volume formula run backwards — and the surface-area formula run backwards too.

LESSON 140–142

3 × 40 MIN

MATEMATIKA 6, §29

S7 15 DISTANCE, AREA AND VOLUME

LESSON CLOCK

25'

30'

35'

25'

5

Reversing the volume formula	25 min
The edge of a cube	30 min
From the surface area	35 min
Depth and height in context	25 min
Homework	5 min

LEARNING OBJECTIVES

- Find the third edge of a cuboid from its volume and two edges.
- Find the edge of a cube from its volume or its surface area.
- Find a missing edge from the total surface area of a cuboid.
- Solve depth-of-water and height-of-box problems in the right units.

TEXTBOOK

National	pp. 404–413
Cambridge	Stage 7, pp. 156–162
Workbook	Exercise 15.2

01

Explanation

Reversing the volume formula

$V = abc$ is a multiplication of three numbers. If two of them and the product are known, the third is a division.

$$c = \frac{V}{ab}$$

VOLUME	BASE	BASE AREA	MISSING EDGE
120 cm^3	5×4	20 cm^2	6 cm
240 cm^3	10×6	60 cm^2	4 cm
84 cm^3	7×3	21 cm^2	4 cm
150 cm^3	5×5	25 cm^2	6 cm

MULTIPLY THE TWO YOU HAVE, THEN DIVIDE ONCE

Finding the base area first turns a three-number problem into a two-number one. It also makes the units obvious: $cm^3 \div cm^2 = cm$.

The edge of a cube

For a cube all three edges are the same, so $V = a^3$ and the edge is the number whose cube is the volume.

VOLUME	QUESTION ASKED	EDGE
27 cm^3	$a^3 = 27$	3 cm
64 cm^3	$a^3 = 64$	4 cm
343 cm^3	$a^3 = 343$	7 cm
512 cm^3	$a^3 = 512$	8 cm
1000 cm^3	$a^3 = 1000$	10 cm

DO NOT DIVIDE THE VOLUME BY THREE

a^3 means $a \cdot a \cdot a$, not $3a$. A cube of volume 27 has edge 3, not 9 — and $3 \cdot 3 \cdot 3$ checks it in one line.

Keeping the first ten cubes in mind — 1, 8, 27, 64, 125, 216, 343, 512, 729, 1000 — makes almost every question of this kind immediate.

From the surface area

For a cube, $S = 6a^2$: divide by six, then take the square root.

SURFACE AREA	A^2	EDGE
54 cm^2	9	3 cm
96 cm^2	16	4 cm
150 cm^2	25	5 cm
216 cm^2	36	6 cm

For a cuboid with two edges known the same idea gives an equation instead of a division. With $a = 5$, $b = 4$ and $S = 94$:

$2(20 + 4c + 5c) = 94$, so $40 + 18c = 94$, $18c = 54$, $c = 3$

TWO EDGES	SURFACE AREA	EQUATION	THIRD EDGE
5 and 4	94 cm ²	40 + 18c = 94	3 cm
8 and 5	132 cm ²	80 + 26c = 132	2 cm
10 and 6	248 cm ²	120 + 32c = 248	4 cm
6 and 4	108 cm ²	48 + 20c = 108	3 cm

THE FACE YOU KNOW IS A CONSTANT, THE OTHER TWO ARE THE UNKNOWN

Only the pair of faces built from the two known edges contributes a fixed number. The other four faces all carry c , which is why the equation is always linear.

Depth and height in context

Almost every practical question is the volume formula reversed, with a unit conversion in front of it.

QUESTION	VOLUME IN ONE UNIT	BASE AREA	ANSWER
a tank 80 × 50 cm holding 120 litres: its depth	120 000 cm ³	4000 cm ²	30 cm
a pool 25 × 10 m holding 500 m ³ : its depth	500 m ³	250 m ²	2 m
a box of 360 cm ³ on a 12 × 6 base: its height	360 cm ³	72 cm ²	5 cm
a brick of 1920 cm ³ , 20 long and 12 wide	1920 cm ³	240 cm ²	8 cm

LITRES ARE NOT LENGTHS

A capacity in litres must become cm³ — multiply by 1000 — before it can be divided by an area in cm². Dividing 120 by 4000 gives 0.03, which is the right arithmetic on the wrong numbers.

02 Worked examples

EXAMPLE 1 A cuboid has volume 240 cm^3 and a base 10 cm by 6 cm . Find its height.

① Base area $10 \cdot 6 = 60 \text{ cm}^2$.

② $240 \div 60$.

③ $= 4 \text{ cm}$.

Check: $10 \cdot 6 \cdot 4 = 240 \checkmark$

ANSWER

4 cm

EXAMPLE 2 A cube has volume 512 cm^3 . Find its edge.

① We need a with $a^3 = 512$.

② $8 \cdot 8 = 64$ and $64 \cdot 8 = 512$.

③ $a = 8 \text{ cm}$.

Not $512 \div 3$ — the edges multiply, they do not add.

ANSWER

8 cm

EXAMPLE 3 A cuboid is 5 cm by 4 cm and has surface area 94 cm². Find the third edge.

① $S = 2(5 \cdot 4 + 4c + 5c) = 94.$

② $40 + 18c = 94$, so $18c = 54.$

③ $c = 3$ cm.

Check: $2(20 + 12 + 15) = 94$ ✓

ANSWER

3 cm

03 Interactive model

Give the volume and one pair of edges and let the class find the third by trial before the division is written down; the division then looks like a shortcut, not a rule.

Working the formula backwards

A cuboid of volume 120 cm^3 on a 5×4 base has height:

A cuboid of volume 84 cm^3 on a 7×3 base has height:

A cube of volume 343 cm^3 has edge:

A cube of volume 27 cm^3 has edge:

A cube of surface area 150 cm^2 has edge:

A cuboid 8 by 5 with $S = 132 \text{ cm}^2$ has third edge:

A tank $80 \times 50 \text{ cm}$ holding 120 litres is deep:

To find a missing edge from the volume you:

divide by 3

divide by the base area

take the square root

multiply

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Missing edge	Noma'lum qirra	Неизвестное ребро
Base	Asos	Основание
Height	Balandlik	Высота
Depth	Chuqurlik	Глубина
Base area	Asos yuzasi	Площадь основания
Cube root	Kub ildiz	Кубический корень
Capacity	Sig'im	Вместимость
To rearrange	O'zgartirib yozish	Выразить

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

From $V = abc$, the edge c equals:

A cuboid of volume 240 cm^3 on a 10×6 base is:

A cube of volume 1000 cm^3 has edge:

A cube of surface area 96 cm^2 has edge:

16

4

6

8

A capacity in litres must first be turned into:

cm^2

cm^3

metres

nothing

The units of volume divided by area are:

cm^3

cm^2

cm

litres

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $V = 120 \text{ cm}^3$ on a 5×4 base: the height

ANSWER 6 cm

2. $V = 240 \text{ cm}^3$ on a 10×6 base: the height

ANSWER 4 cm

3. $V = 84 \text{ cm}^3$ on a 7×3 base: the height

ANSWER 4 cm

4. A cube of volume 27 cm^3 : the edge

ANSWER 3 cm

5. A cube of volume 343 cm^3 : the edge

ANSWER 7 cm

6. A cube of surface area 96 cm^2 : the edge

ANSWER 4 cm

7. From $V = abc$ the third edge is

ANSWER $V \div (ab)$

Medium · the standard question

7 PROBLEMS

1. A cube of volume 512 cm^3 : the edge

ANSWER 8 cm

2. A cube of surface area 150 cm^2 : the edge

ANSWER 5 cm

3. $V = 150 \text{ cm}^3$ on a 5×5 base: the height

ANSWER 6 cm

4. A box of 360 cm^3 on a 12×6 base: the height

ANSWER 5 cm

5. A brick of 1920 cm^3 , 20 cm by 12 cm: the height

ANSWER 8 cm

6. A cube of surface area 54 cm^2 : the edge

ANSWER 3 cm

7. A cube of volume 1000 cm^3 : the edge

ANSWER 10 cm

Hard · stretch and reason

7 PROBLEMS

1. A cuboid 5 by 4 with $S = 94 \text{ cm}^2$: the third edge

ANSWER 3 cm

2. A cuboid 8 by 5 with $S = 132 \text{ cm}^2$: the third edge

ANSWER 2 cm

3. A cuboid 10 by 6 with $S = 248 \text{ cm}^2$: the third edge

ANSWER 4 cm

4. A tank $80 \times 50 \text{ cm}$ holding 120 litres: the depth

ANSWER 30 cm

5. A pool 25 m by 10 m holding 500 m^3 : the depth

ANSWER 2 m

6. A cuboid 12 by 3 has the volume of a cube of edge 6: its third edge

ANSWER 6 cm

7. A cuboid 6 by 4 with $S = 108 \text{ cm}^2$: its volume

ANSWER 72 cm^3

07 Homework

Homework — 5 tasks

Check every answer by putting the edge back into the formula it came from.

1. A cuboid has volume 180 cm^3 and a base 9 cm by 5 cm. Find its height.
2. A cube has volume 729 cm^3 . Find its edge.
3. A cube has surface area 294 cm^2 . Find its edge.
4. A tank with base 60 cm by 40 cm holds 96 litres. Find its depth.
5. A cuboid is 7 cm by 3 cm and has surface area 122 cm^2 . Find the third edge.